2.1 Analog and digital data

Data can be Analog or Digital.

Analog data is continuous information, such as sounds produced by a human voice.

Information with discrete states is referred to as digital data. Discrete values are assigned to digital data.

In contrast to digital data, which has discrete values, analogue data is represented in a continuous form.

Data that is saved in a format that resembles its original structure. Analogue signals, in contrast to digital data, recreate data by using constantly varying electric currents and voltages.

It's difficult to remove noise and wave distortions during transmission in an analogue system since data is conveyed using fluctuating currents. As a result, analogue transmissions are incapable of high-quality data transfer.

Digital signals, on the other hand, use binary data strings (0 and 1) to duplicate the data being conveyed. Noise and distortions are minimal, allowing for high-quality data transmission.

Analog and digital storage are two methods for storing data or information. The data must be in distinct digital form for a computer to use it. Signals, like data, can take both analogue and digital forms. Data must first be transformed to digital form before it can be transmitted digitally.

2.2 Analog and digital signals

The way data is transmitted through the medium is called signaling. It communicates through the use of electricity.

Types of signaling

1. Digital signaling
2. Analog signaling

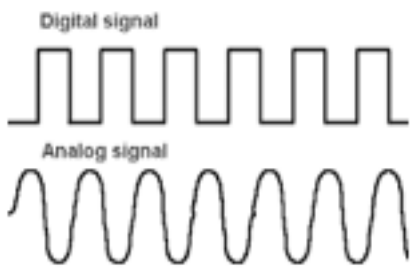


Fig: Types of signaling

Digital signaling

Digital signaling is used in the majority of computer networks.

Encoding schemes are used to encode data in a digital signal.

- Current state encoding
- State transition encoding

Current state encoding

- The presence or absence of a signal characteristic or state is used to encrypt data in current-state encoding strategies.
- The network monitors the signal on a regular basis.
- The state-transition encoding approach represents data by using signal transitions.
- a 1 indicates the presence of a transition, while a 0 indicates the absence of a transition.

State transition encoding schemes

- Bipolar-Alternate Mark Inversion (AMI)
- Non-Return-to-Zero (NRZ)Manchester
- Differential Manchester
- Biphase Space (FM-0)
- Comparing Signaling Methods

Advantages of digital signaling

- Noise and interruption errors are reduced.
- Less costly equipment is used.

Disadvantages of digital signaling

- Attenuation is a problem that many people have.

Analog signaling

To send the digital data over an analog media, it needs to be converted into analog signal. There can be two cases according to data formatting.

Advantages of analog signaling

- Signal attenuation is lower than that of a digital signal.
- To maximize bandwidth, it can be multiplexed.

Disadvantages of analog signaling

- Noise and interference make it more vulnerable to mistakes.

Encoding

Encoding is the process of translating data, or a given sequence of characters, symbols, alphabets, or other characters, into a specific format for safe data transmission. Decoding is the method of extracting information from a transformed format in the opposite direction of encoding.

Encoding is the method of representing 1s and 0s in digital signals on a transmission connection using different patterns of voltage or current levels.

Unipolar, Polar, Bipolar, and Manchester are the most common forms of line encoding.

Encoding technique

Depending on the method of data conversion, data encoding techniques are classified as follows.

- Analog data to analog signal: This group includes analog signal modulation techniques such as Amplitude Modulation, Frequency Modulation, and Phase Modulation.

- Analog data to digital signal: Digitization is the term for this operation, which is carried out using Pulse Code Modulation PCM. As a result, it's just optical modulation. As previously mentioned, sampling and quantization are critical components in this process. The performance of Delta Modulation is superior to that of PCM.

- Digital data to analog signal: Amplitude Shift Keying ASK, Frequency Shift Keying FSK, Phase Shift Keying PSK, and other modulation techniques fall into this group.

- Digital data to digital signal: There are several ways to map digital data to digital signals.

Difference between analog and digital

Analog	**Digital**
An analog signal is a continuous signal that represents physical measurements.	Digital signals are time separated signals which are generated using digital modulation.
It is denoted by sine waves	It is denoted by square waves
It uses a continuous range of values that help you to represent information.	Digital signal uses discrete 0 and 1 to represent information.
Temperature sensors, FM radio signals, Photocells, Light sensor, Resistive touch screen are examples of Analog signals.	Computers, CDs, DVDs are some examples of Digital signal.
The analog signal bandwidth is low	The digital signal bandwidth is high.

Analog signals are deteriorated by noise throughout transmission as well as write/read cycle.	Relatively a noise-immune system without deterioration during the transmission process and write/read cycle.
Analog hardware never offers flexible implementation.	Digital hardware offers flexibility in implementation.
It is suited for audio and video transmission.	It is suited for Computing and digital electronics.
Processing can be done in real-time and consumes lesser bandwidth compared to a digital signal.	It never gives a guarantee that digital signal processing can be performed in real time.
Analog instruments usually have s scale which is cramped at lower end and gives considerable observational errors.	Digital instruments never cause any kind of observational errors.
Analog signal doesn't offer any fixed range.	Digital signal has a finite number, i.e., 0 and 1.

2.3 Transmission modes: serial and parallel transmission, Asynchronous and synchronous transmission

How data is sent between two computers is determined by the transmission method. Binary data in the form of 1s and 0s can be transmitted in two ways: parallel and serial.

Parallel Transmission

The binary bits are grouped into fixed-length groups. With the same number of data lines, the transmitter and receiver are connected in parallel. Both computers can tell the difference between high and low order data lines. On all lines, the sender sends all the bits at the same time. A whole group of bits (data frame) is transferred in one go because the data lines are equal to the number of bits in a group or data frame.

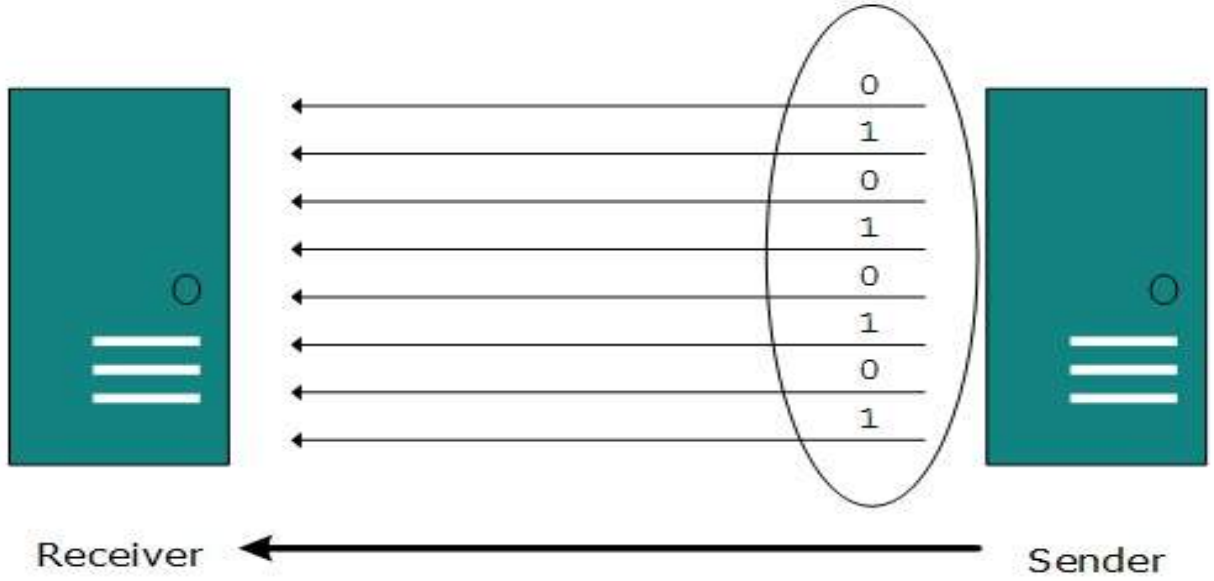


Fig: Parallel transmission

Serial Transmission

Bits are sent one after the other in a queue in serial transmission. Only one communication channel is required for serial transmission.

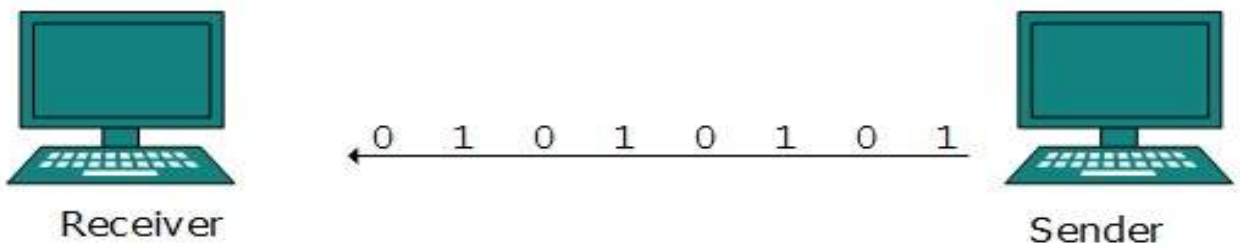


Fig: Serial transmission

Asynchronous or synchronous serial transmission is possible.

Asynchronous Serial Transmission

It's called that since timing isn't important. Data bits have a distinct pattern that aids the receiver in identifying the start and end of data bits. Every data byte, for example, is prefixed with a 0 and one or more 1s are added at the end.

A gap may exist between two continuous data-frames (bytes).

Synchronous Serial Transmission

Because there is no method to detect start and finish data bits in synchronous transmission, timing is critical. There is no prefix/suffix technique or pattern. Burst mode sends data bits without keeping a space between bytes (8-bits). Several bytes can be contained in a single burst of data bits. As a result, timing becomes extremely essential.

The receiver is responsible for recognising and separating bits into bytes.

Synchronous transmission offers the advantage of high speed and does not have the overhead of extra header and footer bits that asynchronous transmission has.

2.4 Communication media: guided media and unguided

Physical transmission media used in communications include twisted-pair cable, coaxial cable, and fiber-optic cable. These cables typically are used within or underground between buildings. Ethernet and token ring LANs often use physical transmission media.

Guided media

The physical medium by which signals are transmitted is referred to as the transmission medium. Bounded media is another name for it.

Types of guided media:

Twisted-Pair Cable

One of the more commonly used transmission media for network cabling and telephone systems is twisted-pair cable. Each twisted-pair wire consists of two separate insulated copper wires that are twisted together. The wires are twisted together to reduce noise. Noise is an electric- cal disturbance that can degrade communications.

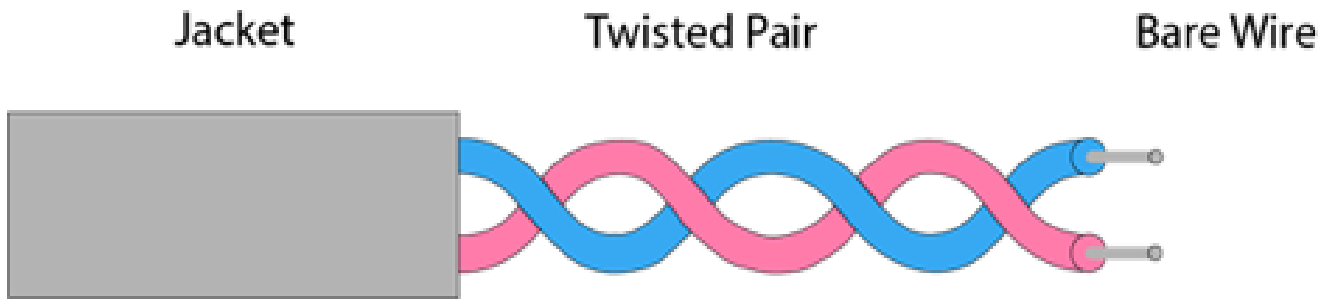


Fig: Twisted pair cable

Types of twisted pair cable

1. Unshielded twisted pair
2. Shielded twisted pair

Unshielded Twisted Pair:

An unshielded twisted pair is widely used in telecommunication. Following are the categories of the unshielded twisted pair cable:

- Category 1: Category 1 is used for telephone lines that have low-speed data.
- Category 2: It can support upto 4Mbps.
- Category 3: It can support upto 16Mbps.
- Category 4: It can support upto 20Mbps. Therefore, it can be used for long-distance communication.
- Category 5: It can support upto 200Mbps.

Advantages

- It is cheap.
- Installation of the unshielded twisted pair is easy.
- It can be used for high-speed LAN.

Disadvantage:

- This cable can only be used for shorter distances because of attenuation.

Shielded Twisted Pair

A shielded twisted pair is a cable that contains the mesh surrounding the wire that allows the higher transmission rate.

Characteristics of Shielded Twisted Pair:

- The cost of the shielded twisted pair cable is not very high and not very low.
- An installation of STP is easy.
- It has higher capacity as compared to unshielded twisted pair cable.
- It has a higher attenuation.
- It is shielded that provides the higher data transmission rate.

Disadvantages

- It is more expensive as compared to UTP and coaxial cable.
- It has a higher attenuation rate.

Coaxial Cable

Coaxial cable, often referred to as coax, consists of a single copper wire surrounded by at least three layers: (1) an insulating material, (2) a woven or braided metal, and (3) a plastic outer coating.

Cable television (CATV) network wiring often uses coaxial cable because it can be cabled over longer distances than twisted-pair cable. Most of today's computer networks, however, do not use coaxial cable because other transmission media such as fiber-optic cable transmit signals at faster rates.

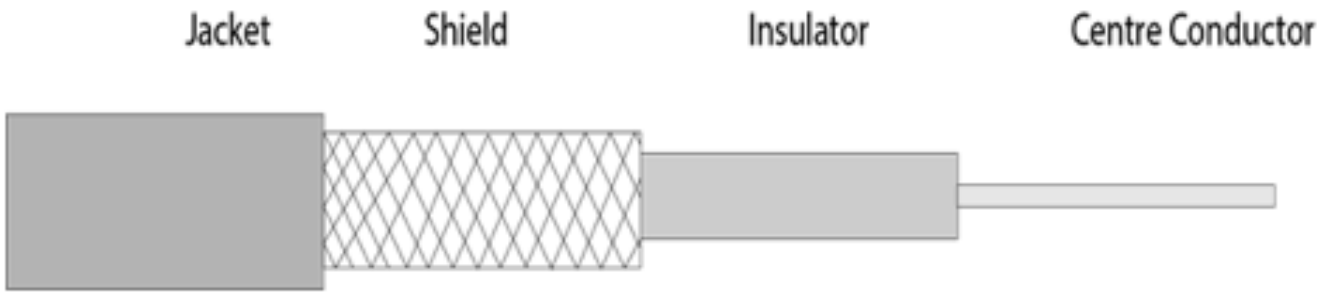


Fig: Coaxial cable

Types of coaxial cable

1. Baseband transmission
2. Broadband transmission

Advantages of Coaxial cable:

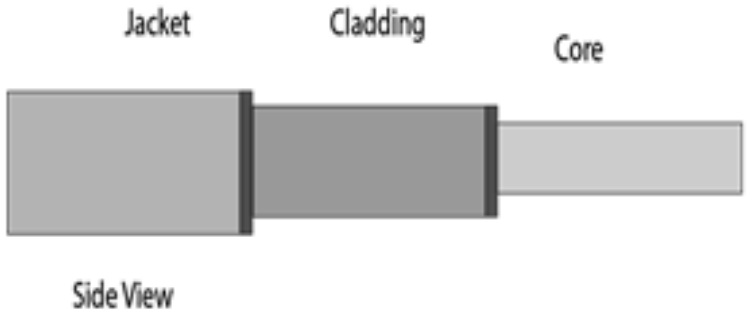
- The data can be transmitted at high speed.
- It has better shielding as compared to twisted pair cable.
- It provides higher bandwidth.

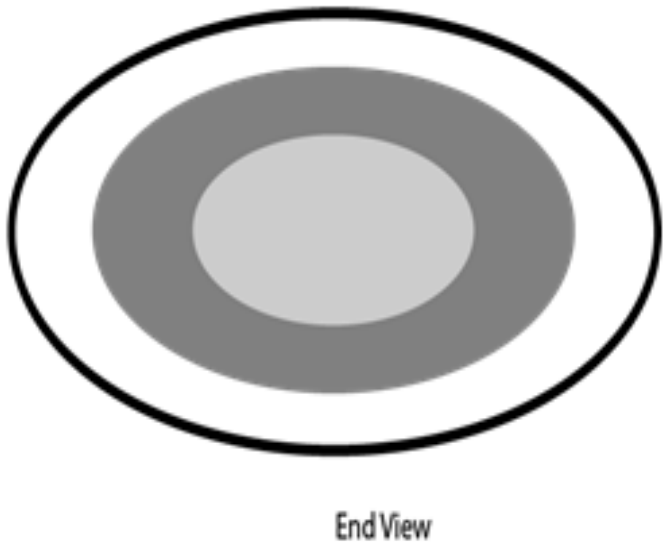
Disadvantages of Coaxial cable:

- It is more expensive as compared to twisted pair cable.
- If any fault occurs in the cable causes the failure in the entire network.

Fiber-Optic Cable

The core of a fiber-optic cable consists of dozens or hundreds of thin strands of glass or plastic that use light to transmit signals. Each strand, called an optical fiber, is as thin as a human hair. Inside the fiber-optic cable, an insu- lating glass cladding and a protective coating surround each optical fiber.





Fiber-optic cables have the following advantages over cables that use wire, such as twisted-pair and coaxial cables:

- Capability of carrying significantly more signals than wire cables
- Faster data transmission
- Less susceptible to noise (interference) from other devices such as a copy machine
- Better security for signals during transmission because they are less susceptible to noise
- Smaller size (much thinner and lighter weight)

Following are the advantages of fiber optic cable over copper:

- Greater Bandwidth: The fiber optic cable provides more bandwidth as compared copper. Therefore, the fiber optic carries more data as compared to copper cable.

- Faster speed: Fiber optic cable carries the data in the form of light. This allows the fiber optic cable to carry the signals at a higher speed.

- Longer distances: The fiber optic cable carries the data at a longer distance as compared to copper cable.

- Better reliability: The fiber optic cable is more reliable than the copper cable as it is immune to any temperature changes while it can cause obstruct in the connectivity of copper cable.

- Thinner and Sturdier: Fiber optic cable is thinner and lighter in weight so it can withstand more pull pressure than copper cable.

Unguided media

The electromagnetic waves are transmitted without the use of any physical medium in an unguided transmission. Therefore it is also known as wireless transmission.

Air is the medium through which electromagnetic energy can easily flow in unguided media.

Unguided transmission can be categorized into three types:

Radio waves: Radio waves are electromagnetic waves that travel through free space in all directions. Radio waves are omnidirectional, which means that the signals travel in all directions. Radio waves have a frequency range of 3 kHz to 1 kHz. The transmitting and receiving antennas are not compatible in the case of radio waves, so the wave transmitted by the sending antenna can be received by any receiving antenna.

Applications of Radio waves:

- A Radio wave is useful for multicasting when there is one sender and many receivers.
- An FM radio, television, cordless phones are examples of a radio wave.

Advantages of Radio transmission:

- Radio transmission is mainly used for wide area networks and mobile cellular phones.
- Radio waves cover a large area, and they can penetrate the walls.
- Radio transmission provides a higher transmission rate.

Infrared: Infrared transmission is a wireless communication technology that works over short distances. Infrared waves with frequencies ranging from 300 GHz to 400 THz. It's used for short-range communication including data transfer between two mobile phones, TV remote control, and data transfer between a computer and a cell phone that's in the same room.

Characteristics of Infrared:

- It supports high bandwidth, and hence the data rate will be very high.
- Infrared waves cannot penetrate the walls. Therefore, the infrared communication in one room cannot be interrupted by the nearby rooms.
- An infrared communication provides better security with minimum interference.

Satellite Microwave Communication

- A satellite is a physical object that revolves around the earth at a known height.
- Satellite communication is more reliable nowadays as it offers more flexibility than cable and fiber optic systems.
- We can communicate with any point on the globe by using satellite communication.

How Does Satellite work?

The satellite accepts the signal that is transmitted from the earth station, and it amplifies the signal. The amplified signal is retransmitted to another earth station.

Advantages of Satellite Microwave Communication:

- The coverage area of a satellite microwave is more than the terrestrial microwave.
- The transmission cost of the satellite is independent of the distance from the centre of the coverage area.
- Satellite communication is used in mobile and wireless communication applications.
- It is easy to install.
- It is used in a wide variety of applications such as weather forecasting, radio/TV signal broadcasting, mobile communication, etc.

Disadvantages of Satellite Microwave Communication:

- Satellite designing and development requires more time and higher cost.
- The Satellite needs to be monitored and controlled on regular periods so that it remains in orbit.
- The life of the satellite is about 12-15 years. Due to this reason, another launch of the satellite has to be planned before it becomes non-functional.